Lecture 8: Support Vector Machines

Machine Learning 2026

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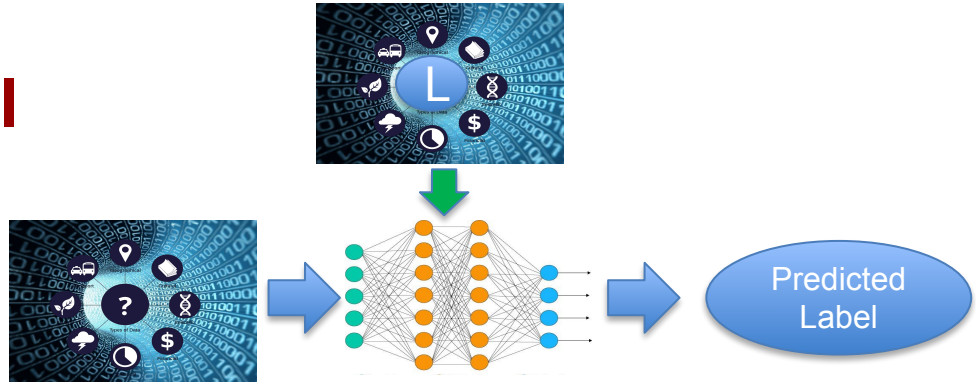
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Recap



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Supervised learning model



- Prediction rule $h : X \rightarrow Y$
  - The learner's output (hypothesis)
  - $A(S)$: hypothesis produced by algorithm A when it is fed training set S
- Data generation model
  - D is a distribution over X **(unknown to the machine learning algorithm)**
  - Instances are labeled according to $f : X \rightarrow Y$ **(unknown to the ML algorithm)**
  - Training set: sampling according to $D, y_i = f(x_i)\ \forall x_i \in S$
- Success metric
  - Classifier error: probability of predicting the wrong label over D

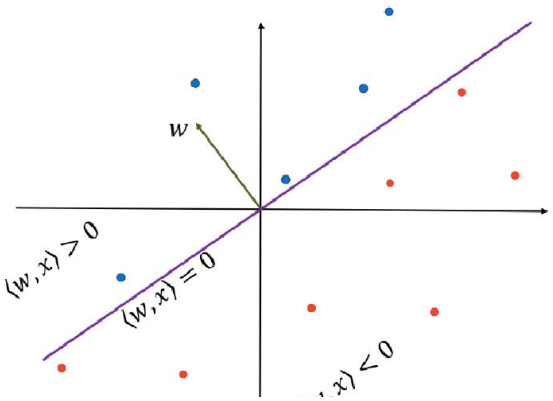
Classification: halfspaces

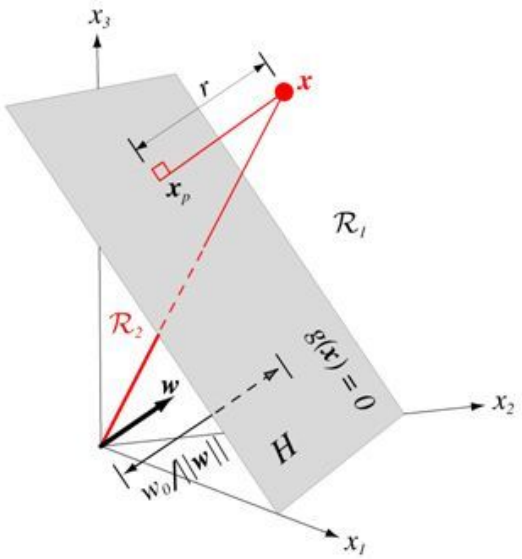
Domain: $X \subset \mathbb{R}^d$

Labels: $Y = \{-1, 1\}$

Loss: 0-1

Hypothesis class: $H = \text{sign} \circ L_d \qquad h_{\mathbf{w}'}(\mathbf{x}') = \text{sign}(\langle \mathbf{w}', \mathbf{x}' \rangle)$

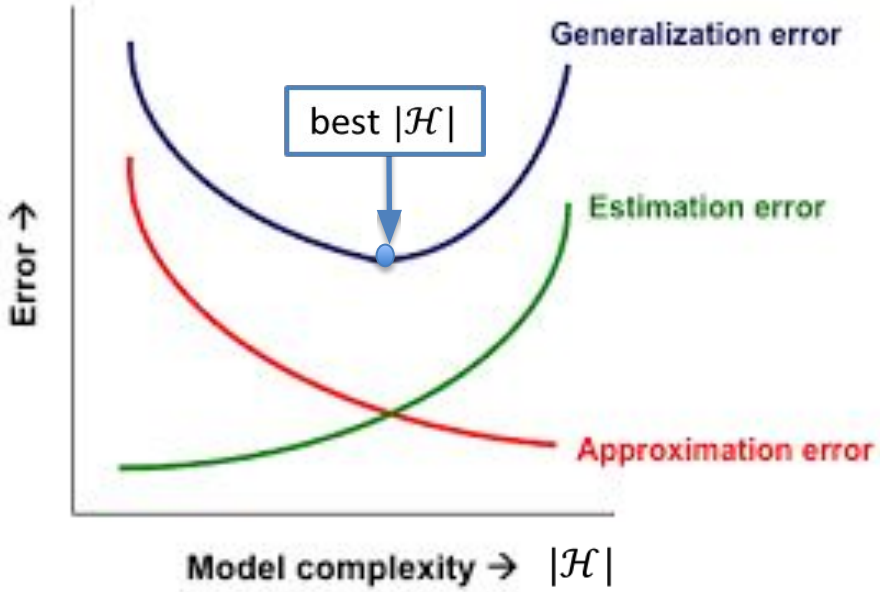




2D: a line divides the two regions (d=2, d'=3)

3D: a plane divides the two regions (d=3, d'=4)

The bias-complexity trade-off



The perceptron algorithm

Start: $\mathbf{w}^{(0)} = (0, \ldots, 0)$

Repeat until convergence:

1. Find a sample with $\text{sign}(\langle \mathbf{w}^{(i)}, \mathbf{x}_m \rangle) = -y_m$
2. Compute the new $\mathbf{w}^{(i+1)} = \mathbf{w}^{(i)} + \alpha \mathbf{x}_m y_m$

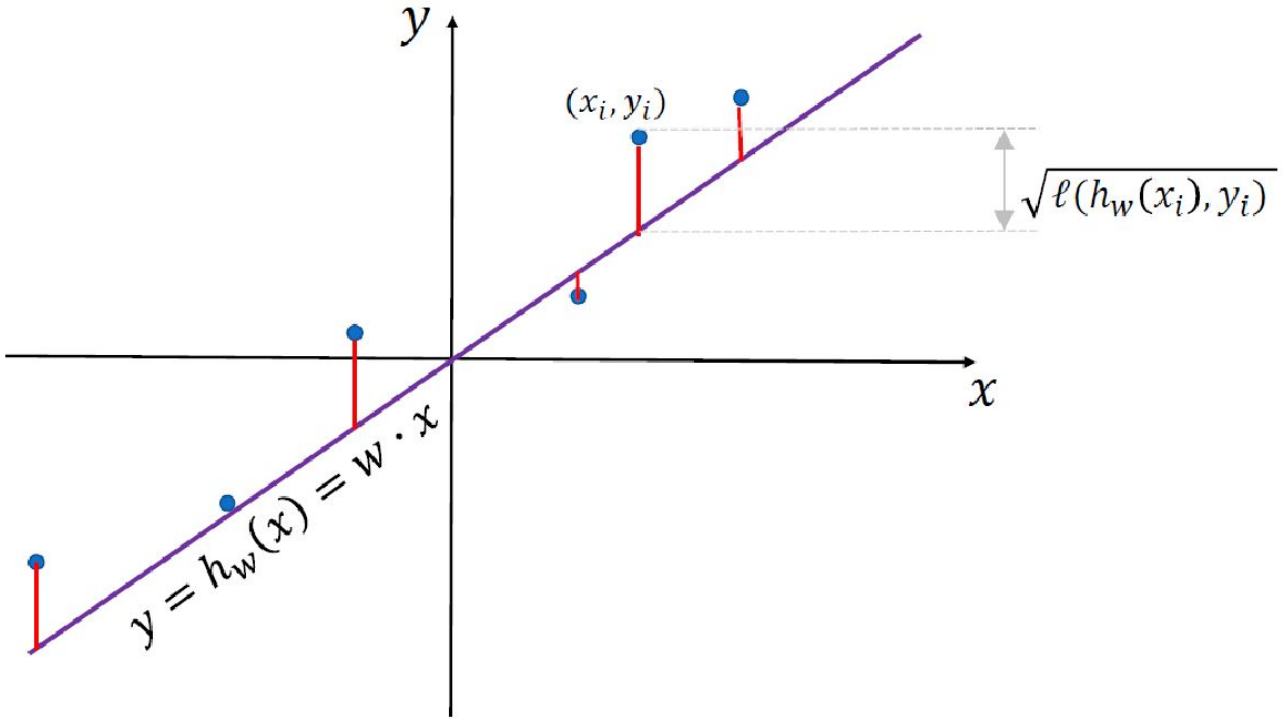
Parameter α is the learning rate (if omitted, it is set to 1)

Termination depends on realizability: if the training set is not linearly separable, the perceptron will go on forever

Perceptron: implementation notes

- ➔ Normalizing the input can reduce convergence time (the perceptron may swing a lot if one dimension has a much wider dynamic range)
- ➔ Randomizing the selection of the **single** sample to use at each iteration can reduce the risk of unlucky orderings (i.e., a lot of misleading samples in the first positions)
- ➔ **Pocket algorithm**: compute the error for each iteration, keep the best one so that we have an approximate solution if the set is not linearly separable
- ➔ The perceptron will reach the ERM solution, but depending on the training, it may reach *any* ERM solution

The linear regression problem



The least squares algorithm

$$\mathbf{A} = \sum_{i=1}^{m} \mathbf{x}_i \mathbf{x}_i^T = \begin{pmatrix} \vdots & & \vdots \\ \mathbf{x}_1 & \dots & \mathbf{x}_m \\ \vdots & & \vdots \end{pmatrix}$$

$$\mathbf{b} = \sum_{i=1}^{m} y_i \mathbf{x}_i = \begin{pmatrix} \vdots & & \vdots \\ \mathbf{x}_1 & \dots & \mathbf{x}_m \\ \vdots & & \vdots \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

There are several solvers for linear systems (e.g., numpy.linalg.solve in Python)

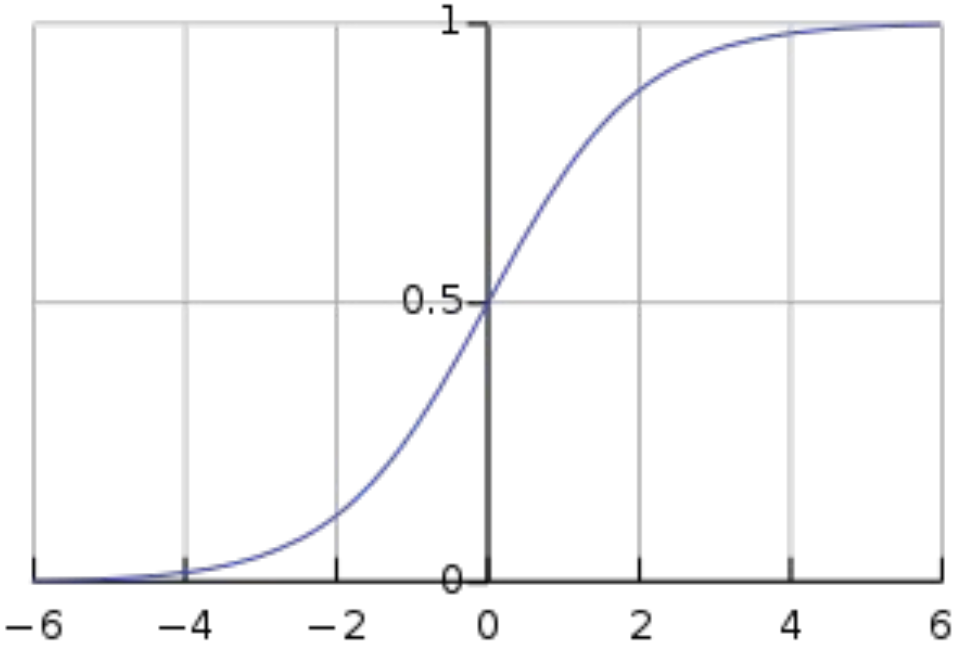
$\mathbf{w} = \mathbf{A}^{-1}\mathbf{b}$

If A is not invertible, we need to handle things a bit differently (see Appendix C in the book)

Logistic regression

We assume that the probability follows a sigmoid function

$$\phi_{\text{sig}}(x) = \frac{1}{1 + e^{-x}}$$



Probability is higher than 0.5 for positive values - this is like a smooth sign function with a soft boundary

Logistic regression

The hypothesis class for logistic regression combines the sigmoid with a linear model

$$H_{\text{sig}} = \phi_{\text{sig}} \circ L_d = \{\mathbf{x} \to \phi_{\text{sig}}(\langle \mathbf{w}, \mathbf{x} \rangle),\ \mathbf{w} \in \mathbb{R}^d\} \qquad h_{\mathbf{w}}(\mathbf{x}) = \frac{1}{1 + e^{-\langle \mathbf{w}, \mathbf{x} \rangle}}$$

We consider a logarithmic loss function

$$\ell(h_{\mathbf{w}}, (\mathbf{x}, y)) = \log\left(1 + e^{-y\langle \mathbf{w}, \mathbf{x} \rangle}\right)$$

If $y\langle \mathbf{w}, \mathbf{x} \rangle > 0$, the exponential is between 0 and 1, and the loss is low. On the other hand, the loss grows approximately linearly as the exponent increases

Part 1: Optimization and gradient descent



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Optimization

Optimization is a field in engineering and computer science concerned with finding the parameters that minimize (or maximize) a given function. An optimization problem is defined by:

- An *objective function*
- A set of *constraints* (expressed as equalities or inequalities)

A valid solution needs to meet all constraints

Convex programming

A problem is convex if we have two conditions:

- The objective function is convex
- The feasible set is convex
  - Inequality constraints should be convex
  - Equality constraints should be affine functions

We can solve convex problems efficiently!

Primal and dual problems

We can define a Lagrangian dual problem by turning a constrained problem into an unconstrained one.

Each constraint becomes a penalty in the objective function, with an associated Lagrange multiplier

The solution to the dual problem is *better* than the solution to the primal problem (but it might not be valid)

Example: primal and dual problems

Problem: minimize $f(x, y) = (x + y)^2$ such that $x^2 + y^2 \leq 1$

We move to the unconstrained problem

$$f'(x, y) = (x + y)^2 + \lambda(x^2 + y^2 - 1)$$

This is still a convex function! We can find the minimum by setting the gradient to 0. By gradually increasing the multiplier, we get closer and closer to the primal problem

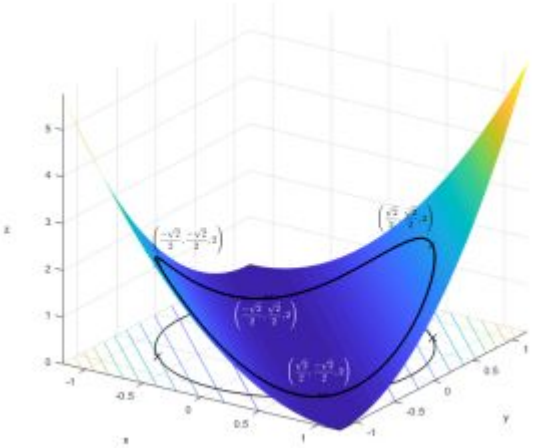


Image: Wikimedia Foundation

Using the gradient

Since we have a single minimum, how about trying to follow the gradient?

This is something like a ball falling down a curved surface and coming to rest on the lowest point

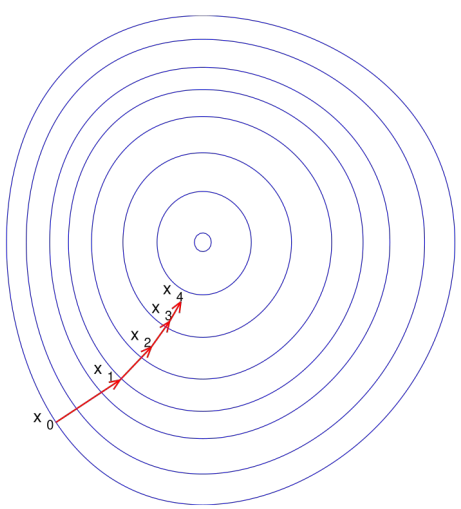


Image: Wikimedia Foundation

Gradient descent

$$\nabla f(\mathbf{w}) = \left(\frac{\partial f}{\partial w_1}, \ldots, \frac{\partial f}{\partial w_d} \right)$$

The gradient points towards the direction of steepest ascent, it corresponds to the first-order Taylor approximation of the function

The approximation works for small steps, but breaks down over larger ones

Gradient descent algorithm

GD algorithm:

$\boldsymbol{w}^{(0)} \leftarrow \mathbf{0}$

for $t \leftarrow 0$ to $T - 1$ do

$$\boldsymbol{w}^{(t+1)} = \boldsymbol{w}^{(t)} - \eta \nabla f(\boldsymbol{w}^{(t)})$$

return $\bar{\boldsymbol{w}} = \boldsymbol{w}^{(T)}$

Learning rate
(controls the size of the steps)

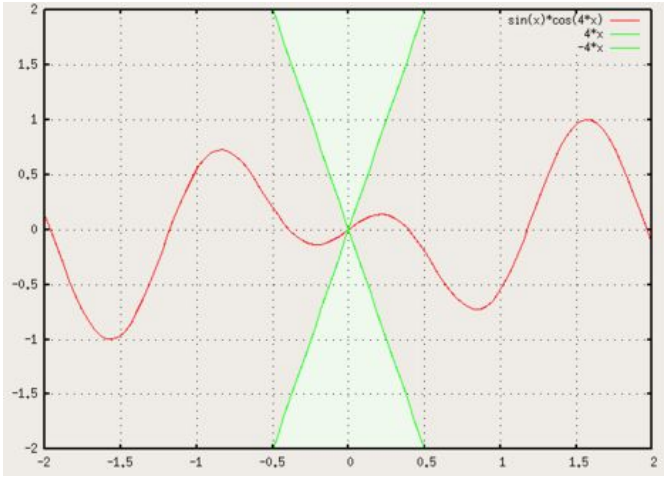
We stop when the gradient is 0 (minimum) or when we reach a maximum number of iterations

Gradient descent: convergence

If the target function is convex and ρ-Lipschitz, and we run GD for T steps, we have:

$$\eta = \sqrt{\frac{B^2}{\rho^2 T}} \Rightarrow f(\bar{\mathbf{w}}) - f(\mathbf{w}^*) \leq \frac{B\rho}{\sqrt{T}}$$

Upper bound on $||\mathbf{w}^*||$



Proof: not part of the course (see Sec. 14.1.1 in the book)

A simple corollary

If the target function is convex and ρ-Lipschitz, and we run GD for T steps, we have:

$$\eta = \sqrt{\frac{B^2}{\rho^2 T}} \Rightarrow f(\bar{\mathbf{w}}) - f(\mathbf{w}^*) \leq \frac{B\rho}{\sqrt{T}}$$

If we want a precision ε, we can run the GD algorithm over enough steps.

$$T \geq \frac{B^2\rho^2}{\varepsilon^2} \Rightarrow f(\bar{\mathbf{w}}) - f(\mathbf{w}^*) \leq \varepsilon$$

This is not learning: the error bound is with probability 1!

Part 2:
Hard SVMs



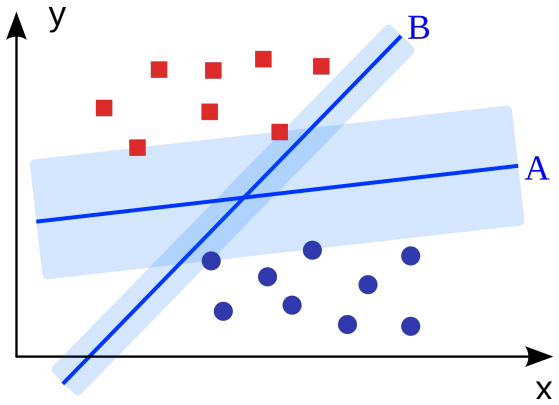
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Margins

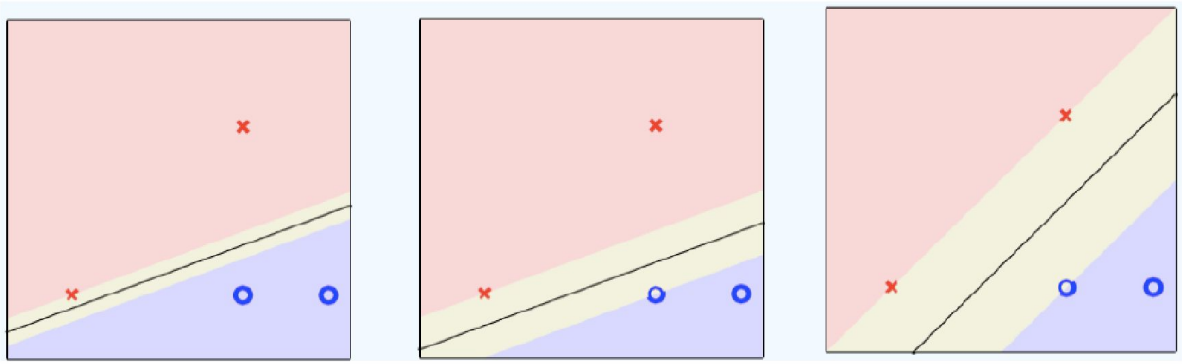
Let us assume realizability (i.e., the data is linearly separable).

We can define the **margin** as the minimum distance between points in the training set and the separating hyperplane

Imagine there is some “noise” on the inputs: a larger margin can tolerate points that are displaced farther



Defining margins



Given a hyperplane defined by $d(\mathbf{x}, L) = \min\{||\mathbf{x} - \mathbf{v}|| : \mathbf{v} \in L\}$ and a point $\mathbf{x}$, the distance between them is

$$L = \{\mathbf{v} : \langle \mathbf{v}, \mathbf{w} \rangle + b = 0\}$$

Distance theorem

If $||\mathbf{w}|| = 1$, we have $d(\mathbf{x}, L) = |\langle \mathbf{w}, \mathbf{x} \rangle + b|$

We can then define the margin as $\min_{\mathbf{x}_i \in S} |\langle \mathbf{w}, \mathbf{x}_i \rangle + b|$

The training set points that are on the margin are called **support vectors**

Proof: first step

The distance between the point and the hyperplane is

$$\min\{||\mathbf{x} - \mathbf{z}||, \mathbf{z} : \langle \mathbf{w}, \mathbf{z} \rangle + b = 0\}$$

Let us define

$$\mathbf{v} = \mathbf{x} - (\langle \mathbf{w}, \mathbf{x} \rangle + b)\mathbf{w}$$

We can easily prove that it lies on the hyperplane:

$$\langle \mathbf{w}, \mathbf{v} \rangle + b = \langle \mathbf{w}, \mathbf{x} \rangle (\langle \mathbf{w}, \mathbf{x} \rangle + b)||\mathbf{w}||^2 + b$$

Since $||\mathbf{w}||^2 = 1$, we have $\langle \mathbf{w}, \mathbf{v} \rangle + b = 0$

Proof: second step

$$\mathbf{v} = \mathbf{x} - (\langle \mathbf{w}, \mathbf{x} \rangle + b)\mathbf{w}$$

The distance between $\mathbf{x}$ and $\mathbf{v}$ is

$$||\mathbf{x} - \mathbf{v}|| = ||\mathbf{x} - \mathbf{x} + (\langle \mathbf{w}, \mathbf{x} \rangle + b)\mathbf{w}||$$

$$||\mathbf{x} - \mathbf{v}|| = |\langle \mathbf{w}, \mathbf{x} \rangle + b|||\mathbf{w}||$$

Since $||\mathbf{w}|| = 1$, we have

$$||\mathbf{x} - \mathbf{v}|| = |\langle \mathbf{w}, \mathbf{x} \rangle + b|$$

Proof: third step

Since $\mathbf{v}$ is on the hyperplane, it is an upper bound to distance. We need to prove that it is the closest point. We can take a generic point on the hyperplane $\mathbf{u}$

$$||\mathbf{x} - \mathbf{u}||^2 = ||(\mathbf{x} - \mathbf{v}) + (\mathbf{v} - \mathbf{u})||^2$$

Expanding the square, we get:

$$||\mathbf{x} - \mathbf{u}||^2 = ||\mathbf{x} - \mathbf{v}||^2 + ||\mathbf{v} - \mathbf{u}||^2 + 2\langle \mathbf{x} - \mathbf{v}, \mathbf{v} - \mathbf{u} \rangle$$

Always positive (it is a norm)

We can then turn this into a bound:

$$||\mathbf{x} - \mathbf{u}||^2 \geq ||\mathbf{x} - \mathbf{v}||^2 + 2\langle \mathbf{x} - \mathbf{v}, \mathbf{v} - \mathbf{u} \rangle$$

Proof: final step

From previous steps, we have:

$$||\mathbf{x}-\mathbf{u}||^2 \geq ||\mathbf{x}-\mathbf{v}||^2 + 2\langle \mathbf{x}-\mathbf{v}, \mathbf{v}-\mathbf{u}\rangle \qquad \mathbf{v} = \mathbf{x} - (\langle \mathbf{w}, \mathbf{x}\rangle + b)\mathbf{w}$$

We can substitute the definition of $\mathbf{v}$ to get:

$$||\mathbf{x}-\mathbf{u}||^2 \geq ||\mathbf{x}-\mathbf{v}||^2 + 2\langle \mathbf{x}-\mathbf{x} + (\langle \mathbf{w}, \mathbf{x}\rangle + b)\mathbf{w}, \mathbf{v}-\mathbf{u}\rangle$$

We exploit commutative property after simplifying the bound

$$||\mathbf{x}-\mathbf{u}||^2 \geq ||\mathbf{x}-\mathbf{v}||^2 + 2(\langle \mathbf{w}, \mathbf{x}\rangle + b)\langle \mathbf{w}, \mathbf{v}-\mathbf{u}\rangle$$

Finally, we note that $\langle \mathbf{w}, \mathbf{u}\rangle = \langle \mathbf{w}, \mathbf{v}\rangle = -b$, as they are both on the hyperplane, so the last term is 0: we simply get that $||\mathbf{x}-\mathbf{u}||^2 \geq ||\mathbf{x}-\mathbf{v}||^2$, which proves our theorem

Hard SVMs

A hard SVM tries to maximize the margin (only on linearly separable data)

Optimization problem:

$$\arg\max_{(\mathbf{w},b):||\mathbf{w}||=1} \min_{\mathbf{x}_i \in S} |\langle \mathbf{w}, \mathbf{x}_i \rangle + b|$$

such that $y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) > 0 \forall (x_i, y_i) \in S$

Equivalent formulation for separable data:

$$\arg\max_{(\mathbf{w},b):||\mathbf{w}||=1} \min_{(x_i,y_i) \in S} y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b)$$

An equivalent convex problem

We can find an equivalent formulation, which is a convex problem:

$$\arg\min_{\mathbf{w},b} ||\mathbf{w}||^2$$

$$\text{such that} \quad y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1$$

This is a convex problem, which we can solve using standard tools (e.g., Lagrange multipliers or GD)

Proof: equivalence of the two problems (first step)

Let us start from the original problem:

$$\arg\max_{(\mathbf{w},b):||\mathbf{w}||=1} \min_{(x_i,y_i)\in S} y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)$$

If $(\mathbf{w}^*, b^*)$ is a solution of the hard SVM problem, the margin is

$$\gamma^* = \min_{(\mathbf{x}_i,y_i)\in S} y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*)$$

If we divide both by the margin, we have

$$y_i\left(\left\langle \frac{\mathbf{w}^*}{\gamma^*}, \mathbf{x}_i \right\rangle + \frac{b^*}{\gamma^*}\right) \geq 1$$

Convex problem constraint!

The optimal hard SVM solution is a valid solution for the convex problem

Proof: equivalence of the two problems (second step)

We now consider the convex problem, whose normalized solution is

$$(\hat{\mathbf{w}}, \hat{b}) = \left(\frac{\mathbf{w}_0}{||\mathbf{w}_0||}, \frac{b_0}{||\mathbf{w}_0||} \right)$$

We know that $||\mathbf{w}_0|| \leq \left|\left| \frac{\mathbf{w}^*}{\gamma^*} \right|\right| = \frac{1}{\gamma^*}$ (or it would not be a valid solution of the convex problem)

The convex problem constraint is $y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1$

$$y_i(\langle \hat{\mathbf{w}}, \mathbf{x}_i \rangle + \hat{b}) = \frac{1}{||\mathbf{w}_0||} y_i(\langle \mathbf{w}_0, \mathbf{x}_i \rangle + b_0) \geq \frac{1}{||\mathbf{w}_0||} \geq \gamma^*$$

The optimal convex solution is an optimal solution of the hard SVM problem!

Computing the margin

The margin is $\gamma^* = \min_{(\mathbf{x}_i, y_i) \in S} y_i(\langle \mathbf{w}^*, \mathbf{x}_i \rangle + b^*)$, and we know that $||\mathbf{w}_0|| = \frac{1}{\gamma^*}$

The margin is then $\gamma^* = \frac{1}{||\mathbf{w}_0||}$

This is consistent with the distance theorem, which states that the margin is the minimum predicted absolute value if the norm of the solution is 1

If $||\mathbf{w}|| = 1$, we have $d(\mathbf{x}, L) = |\langle \mathbf{w}, \mathbf{x} \rangle + b|$

Homogeneous representation

The homogeneous representation is slightly different (bias is included in the normalization), but essentially gets the same results in practice

$$\mathbf{w}_0 = \arg\min_{\mathbf{w}} ||\mathbf{w}||^2$$

$$\text{such that} \quad y_i \langle \mathbf{w}, \mathbf{x}_i \rangle \geq 1 \; \forall (\mathbf{x}_i, y_i) \in S$$

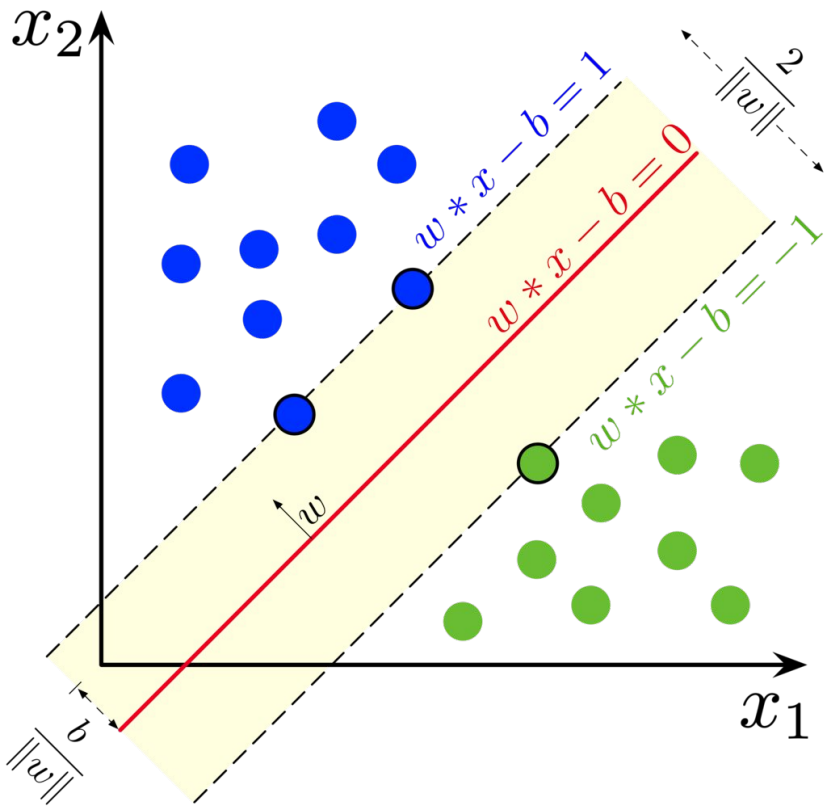
Homogeneous representation

Why is the margin the

Remember the distance theorem:

If $||\mathbf{w}|| = 1$, we have $d(\mathbf{x}, L) = |\langle \mathbf{w}, \mathbf{x} \rangle + b|$

As we are minimizing $||\mathbf{w}||$ instead of fixing it to 1, the margin (i.e., the distance) will scale with its inverse



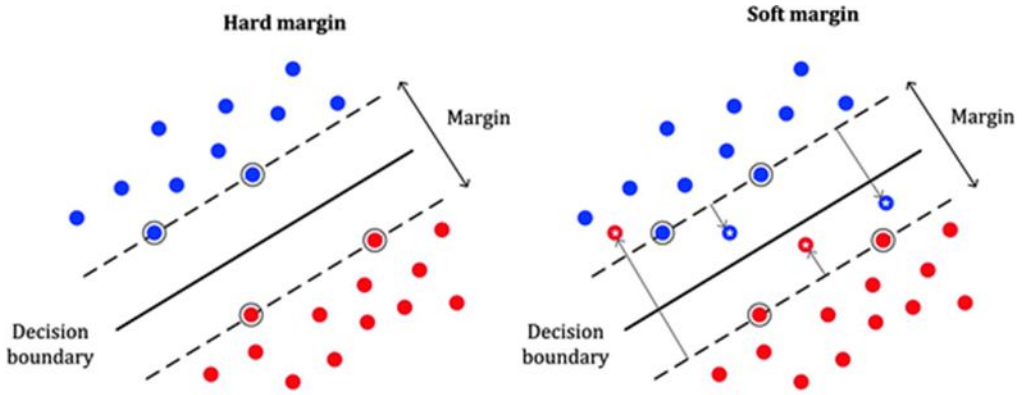
Part 3:
Soft SVMs



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Realizability and soft SVMs

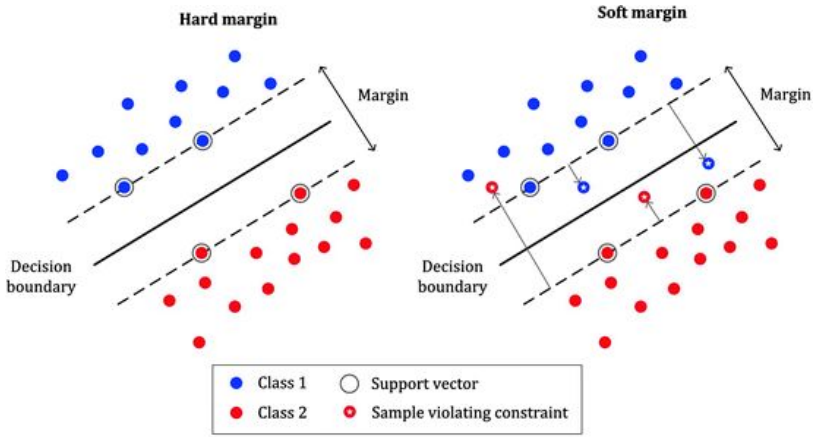
- ➔ Realizability (i.e., linear separability) is a **strong** assumption: even a single outlier breaks the hard SVM model!
- ➔ We may accept a solution in which a few points are misclassified, as long as they are not too many and it is not by a huge margin
- ➔ **Soft SVM**: we define a relaxed version of the problem (as with Lagrange multipliers)



Slack

We define slack variables that indicate how much a point violates the hard SVM constraint:

- ➔ If the point is outside the margin, $\xi_i = 0$
- ➔ If the point is inside the margin but correctly classified, $0 < \xi_i < 1$
- ➔ If the point is misclassified, $\xi_i \geq 1$



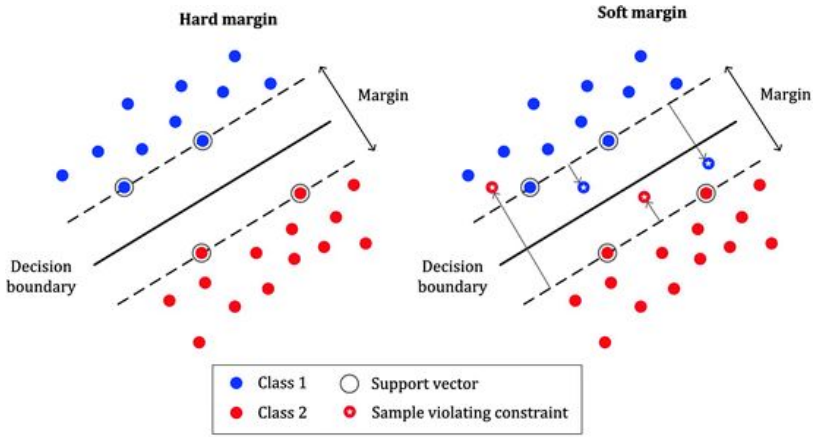
The soft SVM problem

$$\arg\min_{\mathbf{w},b,\xi} \left(\lambda ||\mathbf{w}||^2 + \frac{1}{m}\sum_{i=1}^{m} \xi_i\right)$$

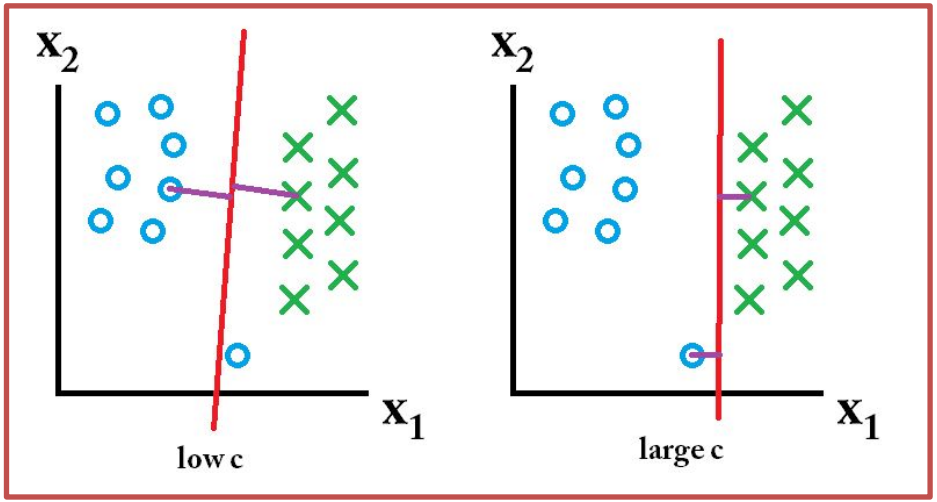
such that: $y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1 - \xi_i \ \forall i$

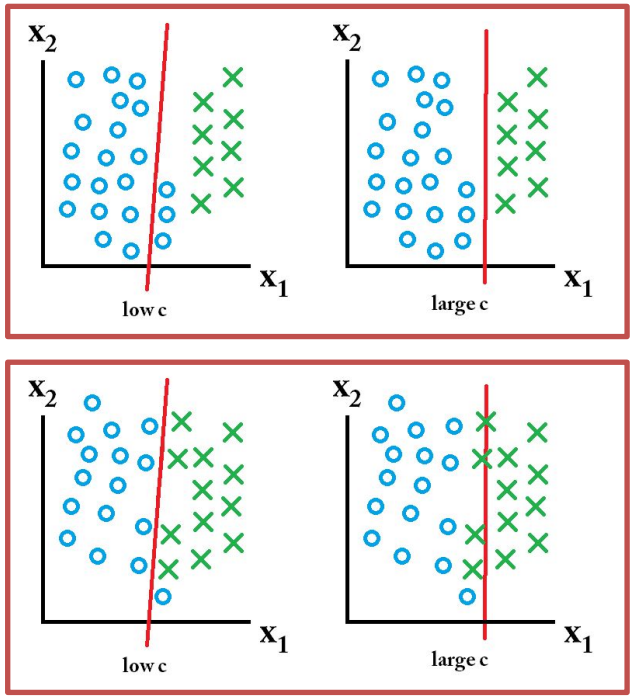
$\xi_i \geq 0 \ \forall i$

Tuning parameter



Finding λ: how soft is too soft?





$c = \lambda^{-1}$ (used by Scikit!)

If $\lambda \to 0$, i.e., $c \to \infty$, we get a hard SVM (like a large Lagrange multiplier)

Soft SVMs and hinge loss

$$\arg\min_{\mathbf{w},b,\xi} \left(\lambda||\mathbf{w}||^2 + \frac{1}{m}\sum_{i=1}^{m}\xi_i\right)$$

such that: $y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \geq 1 - \xi_i \ \forall i$

$\xi_i \geq 0 \ \forall i$

Hinge loss: $\ell^h((\mathbf{w}, b), (\mathbf{x}, y)) = \max\{0, 1 - y_i(\langle \mathbf{w}, \mathbf{x}\rangle + b)$

$$\arg\min_{\mathbf{w},b} \left(\lambda||\mathbf{w}||^2 + \frac{1}{m}\sum_{i=1}^{m}\ell^h((\mathbf{w}, b), (\mathbf{x}_i, y_i))\right)$$

Soft SVMs can be seen as Regularized Loss Minimization (RLM) cases!

The sample complexity of SVMs

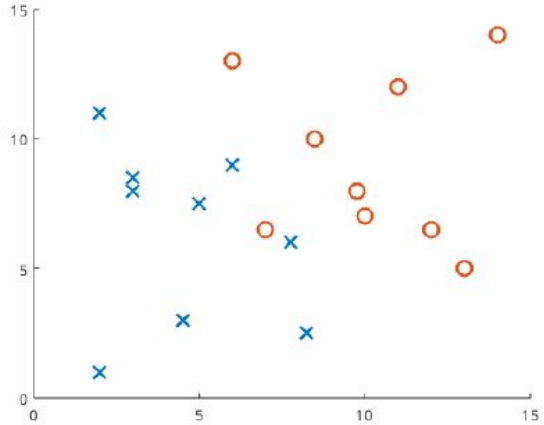
If our distribution D satisfies separability with margin γ and $||\mathbf{x}|| \leq \rho$, with probability $1 - \delta$ the error of a hard SVM with 0-1 loss is bounded by

$$\sqrt{\frac{4\rho^2}{m\gamma^2}} + \sqrt{\frac{2\log(2/\delta)}{m}}$$

If our distribution D has $||\mathbf{x}|| \leq \rho$, with probability $1 - \delta$ the hinge loss of a soft SVM with 0-1 loss is bounded by

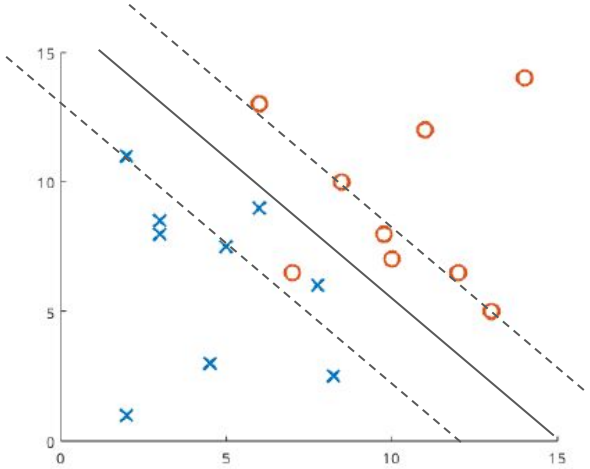
$$L_D[h_{\text{S-SVM}}] \leq L_D(\mathbf{w}^*) + \lambda||\mathbf{w}^*|| + \frac{2\rho^2}{\lambda m}$$

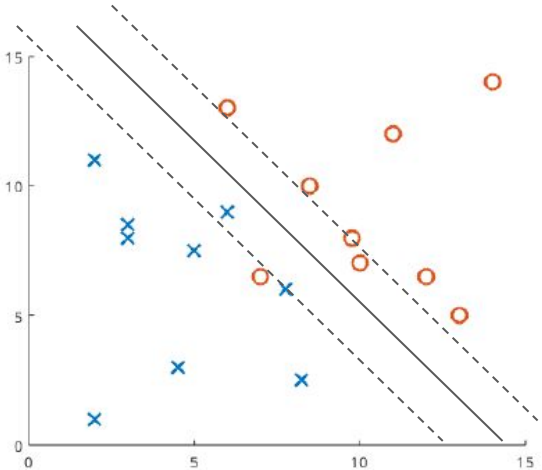
Exercise 2.5



Draw the soft SVM boundary and mark outliers and support vectors

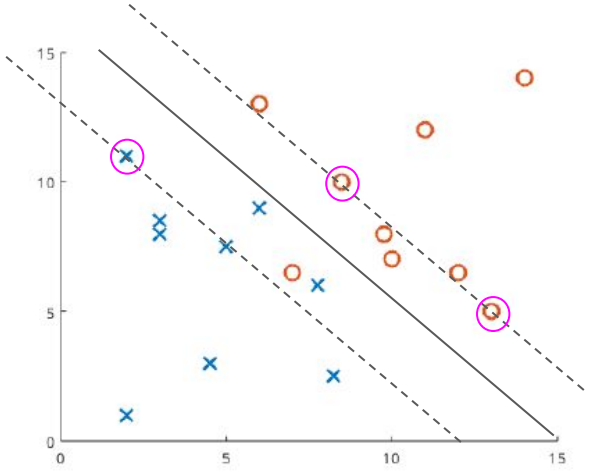
Margin and boundaries

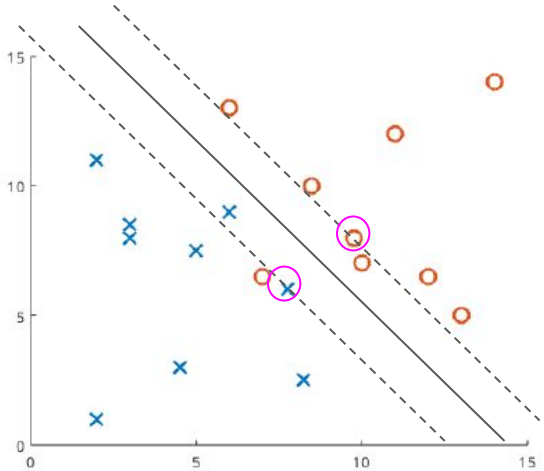




On the left, we selected a relatively large value of λ: we could have been stricter, getting a smaller margin but fewer outliers, as on the right

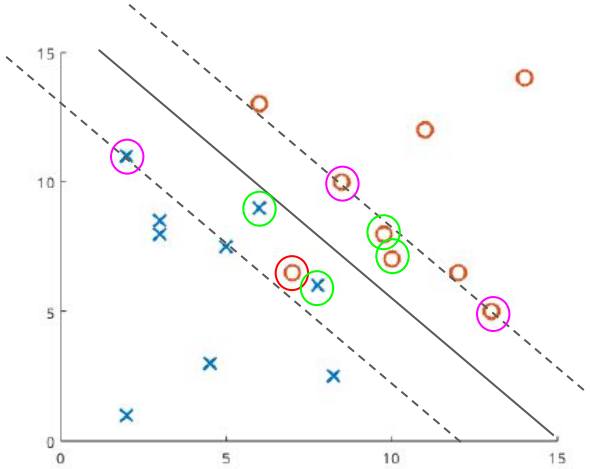
Support vectors





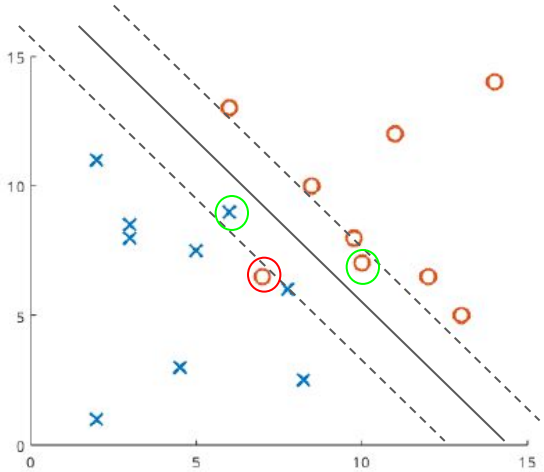
The support vectors are easy to see, as they are on the margin. In the second case, we only have 2 support vectors, which is the minimum possible number (we have to have at least one for each class)

Outliers: large margin



The green circles are outliers with a slack between 0 and 1. The red circle is not only an outlier, but it is also misclassified, with a slack larger than 1 (it looks close to 2 in this case, as it is close to the margin o

Outliers: small margin



In this case, the slack for the two green points is close to 0, as they are close to the margin. On the other hand, the slack for the misclassified point is even larger, and becomes slightly more than 2